
Hayden Austin

Software Engineer

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Software engineer developing React, TypeScript and Firebase applications with staff permissions, reporting and automated release checks. Independent work in GPU programming, recurrent models and numerical validation.

SELECTED ENGINEERING WORK

Equipment catalogue & administration | [Live application](#)

- Built a React/Firebase product catalogue and admin interface with server-side rendering.
- Built consent-gated analytics with batched events, retry handling and scheduled Firestore aggregation for reporting.
- Added CI checks for access rules, server rendering and release integrity, including tests against unauthorised writes and privilege escalation.

Product management & enquiry platform | [Live application](#)

- Built TypeScript spreadsheet import/export and catalogue publishing tools, with compressed image uploads and generated thumbnails.
- Implemented staff provisioning and role-based backend permissions using Firebase Auth custom claims, plus validated, rate-limited enquiry intake.

Independent GPU & ML systems research | [Code & tests](#)

- Implemented recurrent state-space models and parallel scans, integrating custom C++/CUDA and Triton kernels with PyTorch autograd.
- Built tests for outputs, gradients and streaming equivalence against reference implementations, with reproducible benchmarks and artifact-integrity checks.

EXPERIENCE

Freelance Developer

2023 - 2025

- Developed web applications for commerce and professional services, covering product catalogues, authentication, content management and deployment.

Game Server Developer & Administrator

2019 - 2022

- Built event-driven Java plugins with persistent item metadata and server API integrations; maintained Linux servers and investigated performance issues.

TECHNICAL SKILLS

Web: TypeScript, JavaScript, React, Next.js, Node.js, server-side rendering

Backend: Firebase Auth, Firestore, Cloud Functions, SQL

Engineering: Git, GitHub Actions, Linux, automated testing, release verification

Programming & ML: Python, PyTorch, C++, CUDA, Triton, JAX; Java and C#

EDUCATION

Belgium Campus ITversity

Studies toward a Bachelor of Computing, specialising in Software Engineering.